



Lab 5: Build a Database Server

AWS Academy Cloud Foundations — Guided Lab Tutorial

Deploy a fully managed, highly available relational database on Amazon RDS, configure secure networking, and connect a live web application — all within the AWS Management Console.

ESTIMATED TIME: 45-75 MIN

HANDS-ON LAB

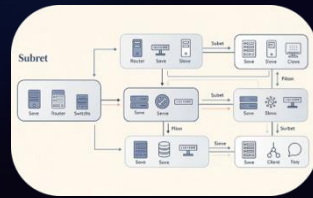
What You'll Build

By the end of this lab, you will have deployed a production-style, Multi-AZ relational database infrastructure entirely within a private VPC — and proven it works by connecting a live web application to it.



Security Group

A dedicated VPC Security Group (DBSG) that allows MySQL traffic *only* from your web server tier — not the open internet.



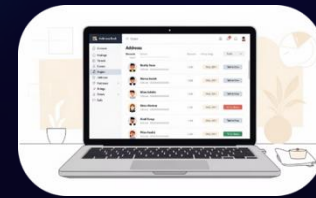
DB Subnet Group

A multi-AZ subnet group spanning two private subnets, giving RDS the placement flexibility it needs for high availability.



RDS MySQL Instance

A fully managed, Multi-AZ MySQL database with automated failover — provisioned without touching a single server.



Web App Integration

An Address Book application that connects to your RDS endpoint, writes records, and persists data across page refreshes.

Learning Objectives

These objectives align with **Bloom's Taxonomy** – moving from applying concrete skills to constructing and validating complete cloud infrastructure.



Apply Security Rules

Configure a custom VPC Security Group that permits database traffic on Port 3306 exclusively from the application tier.



Construct a Subnet Group

Build an Amazon RDS DB Subnet Group spanning multiple Availability Zones inside a VPC.



Provision a Multi-AZ DB

Launch a managed MySQL database instance with automated failover across two Availability Zones.



Validate Connectivity

Link the web application to the RDS endpoint and verify that data persists correctly in the database.

Prerequisites & Environment Profile

What's Already Provisioned for You

When the lab environment starts, AWS Academy automatically provisions the following resources — you do **not** need to create them:

- **Lab VPC** — a pre-configured Virtual Private Cloud with both public and private subnets
- **Web Server (EC2 instance)** — a running instance in the public subnet with the Address Book app pre-installed
- **Web-Security-Group** — the security group already attached to the web server, used as the inbound source for your DB rule

Critical Safety Rules

Stay within AWS Academy guardrails to avoid Access Denied errors:

- Do **NOT** create custom IAM Users or Groups
- Do **NOT** modify root billing settings
- Always select the **LabRole** profile when prompted for IAM identities
- All actions must run under the automated learner credentials provided

Task 1

Accessing the AWS Management Console

Every lab begins with a clean, authenticated console session. Follow these steps carefully before making any configuration changes.

01

Start the Lab

At the top of the lab instructions panel, locate and click the **Start Lab** button. This provisions all pre-built resources in the background.

02

Wait for the Green Light

Watch the status indicator in the upper-left corner. Wait until the circular light next to the **AWS** link turns **solid green** — this confirms the environment is ready.

03

Open the Console

Click the green **AWS** link text. A new browser tab opens and automatically authenticates you into the AWS Management Console — no password needed.

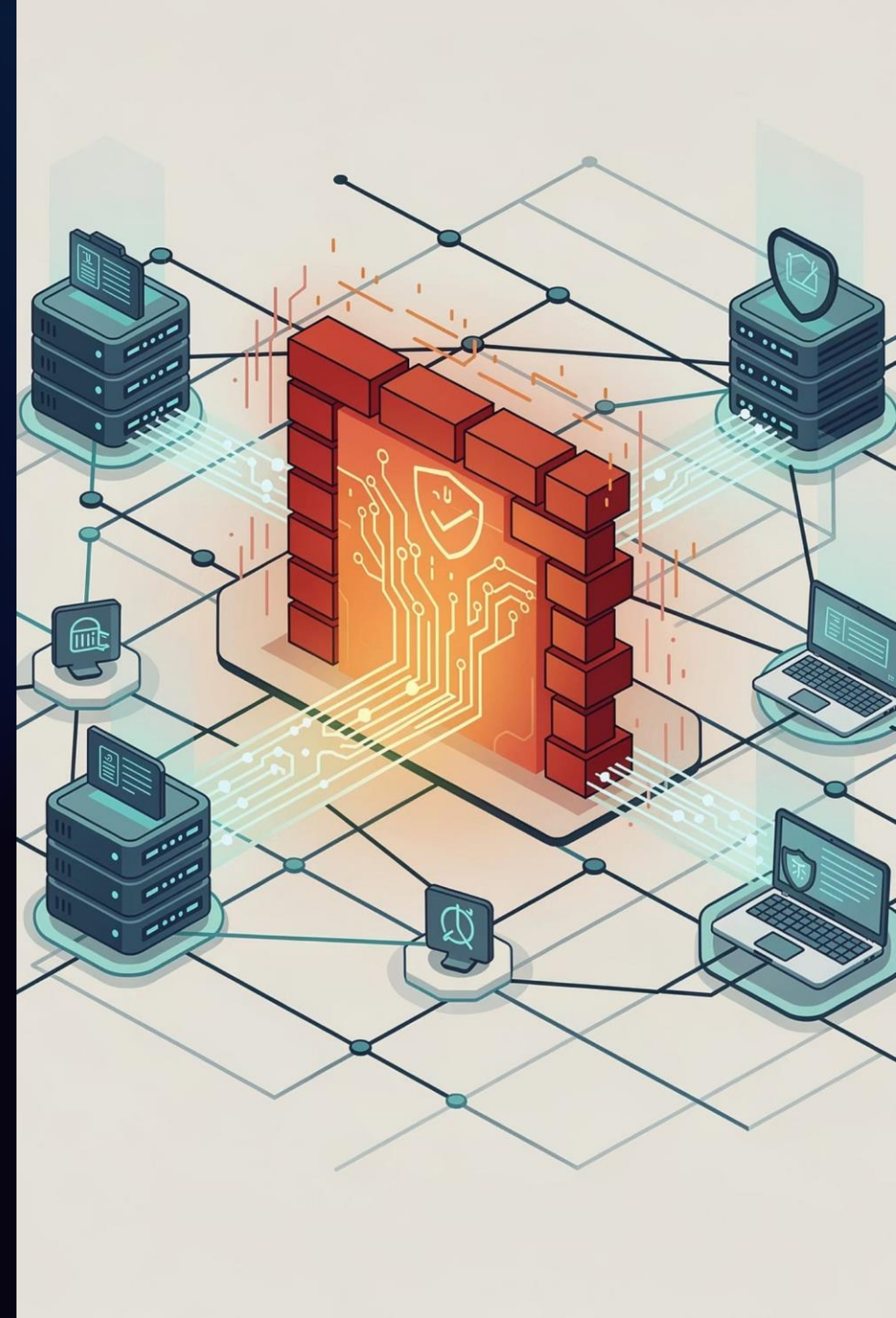


Never proceed past Step 1 until the status light is fully green. Starting tasks too early may result in missing resources.

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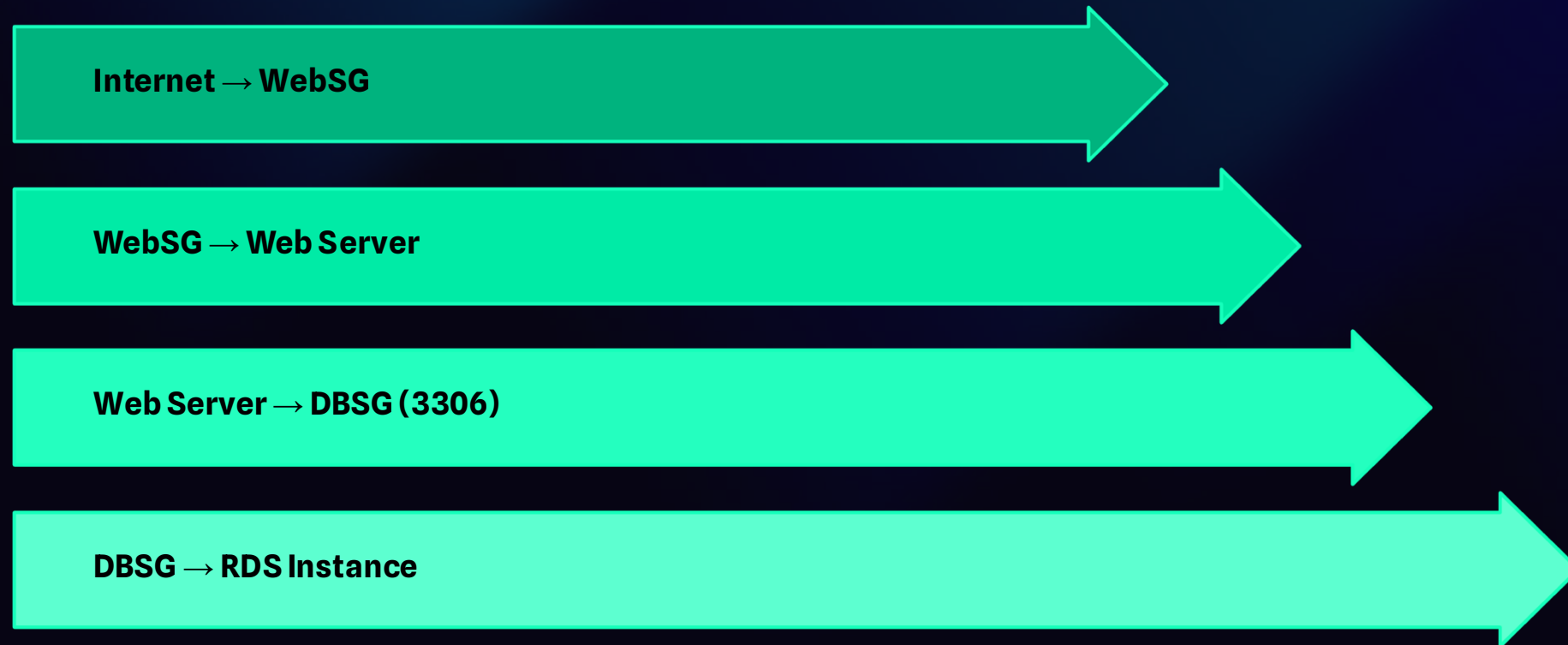
Task 2: Create the DB Security Group (DBSG)

To enforce the **Principle of Least Privilege**, your database must never be directly exposed to the internet. You will create a dedicated Security Group that acts as a virtual firewall — accepting MySQL traffic on Port 3306 *only* from the web server's security group.



Traffic Flow Architecture

Understanding *why* this security design works helps you configure it correctly. Traffic flows in one direction – inbound to the database – and only from trusted sources.



By setting the inbound rule **source** to the `Web-Security-Group` ID (not an IP range), AWS automatically grants access to any instance that belongs to that group – a dynamic, scalable, and secure design pattern.

Security Group Configuration — Step by Step

1 Navigate to VPC Security Groups

In the top global search bar, type **VPC** and select **VPC**. In the left navigation under **Security**, choose **Security groups**. Click the orange **Create security group** button.

2 Fill in Basic Details

Set **Security group name** to **DBSG**, **Description** to **DB Security Group**. For **VPC**, select **Lab VPC** from the dropdown — do **not** leave it on the default VPC.

3 Add the Inbound Rule

Scroll to **Inbound rules** → click **Add rule**. Set **Type** to **MySQL/Aurora (3306)**. Set **Source type** to **Custom**, then search for and select **Web-Security-Group**.

4 Create the Group

Scroll to the bottom of the page and click **Create security group**. Verify the new DBSG appears in the Security Groups list before proceeding.

Task 3: Create a DB Subnet Group

The diagram illustrates a VPC Availability Zone architecture. At the top, a cloud icon represents the Internet. Below it, a large box labeled "Availability Zone" contains three smaller boxes, each representing an Availability Zone. Each Availability Zone box contains a "Subnet" box and a "VPC" box. The "Subnet" boxes are connected to the "VPC" boxes. The "VPC" boxes are connected to the "Internet" cloud. The "VPC" boxes are also connected to each other, forming a mesh topology. The "VPC" boxes are labeled "VPC" and "Subnet". The "Subnet" boxes are labeled "Subnet". The "VPC" boxes are connected to the "Internet" cloud. The "VPC" boxes are also connected to each other, forming a mesh topology. The "VPC" boxes are labeled "VPC" and "Subnet". The "Subnet" boxes are labeled "Subnet".

VPC Availablility Zareoy Zone

DB Subnet Group — Configuration Details

Subnet Group Details

Field	Value to Enter
Name	lab-db-subnet-group
Description	DB Subnet Group for Lab
VPC	Lab VPC

Add Subnets

Setting	Selection
Availability Zones	Both AZs in your lab region (e.g., us-east-1a and us-east-1b)
Subnets	Both private subnets (e.g., 10.0.1.0/24 and 10.0.3.0/24)

Navigate to **RDS** via the global search bar → select **Subnet groups** in the left sidebar → click **Create DB subnet group**. Enter the values above, then click **Create** at the bottom of the form.

 Always select subnets from **Lab VPC only**. Selecting subnets from the default VPC is one of the most common errors in this lab and will break Multi-AZ deployment.

Task 4

Create the Amazon RDS DB Instance

You will now launch a fully managed MySQL database using **Multi-AZ deployment** for high availability. AWS will automatically provision a primary instance and a synchronous standby replica in a second Availability Zone.

Engine

MySQL – Standard Create method

Template

Dev/Test (required for Multi-AZ selection)

Availability

Multi-AZ DB Instance – primary + standby

Instance Class

db.t3.micro (or db.t2.micro)

RDS Settings — Credentials & Storage

Settings (Credentials)

Field	Value
DB instance identifier	lab-db
Master username	main
Master password	lab-password
Confirm password	lab-password

Storage Configuration

Field	Value
Storage type	General Purpose SSD (gp2)
Allocated storage	20 GiB
Storage autoscaling	Uncheck — disable



Credentials are **case-sensitive**. Use exactly **main** and **lab-password** — you will re-enter these in the web application later.

RDS Settings — Connectivity & Additional Config

These connectivity settings ensure your database remains **completely private** — accessible only through the DBSG you created, from within the Lab VPC.

Section	Field	Value / Action
Connectivity	VPC	Lab VPC
Connectivity	DB Subnet group	lab-db-subnet-group
Connectivity	Public access	No — database stays private
Connectivity	VPC security groups	Select DBSG ; remove the default SG by clicking X
Additional configuration	Initial database name	lab
Additional configuration	Automated backups	Uncheck — disable
Additional configuration	Enhanced monitoring	Uncheck — disable

When all settings are confirmed, click the orange **Create database** button. RDS will begin provisioning — this takes **8 to 12 minutes**.

While You Wait: What Is Multi-AZ Doing?

The "Creating" status you see means AWS is doing significant work behind the scenes. Use this time to understand what's being built.



Primary Instance

AWS provisions the main MySQL instance in your first selected Availability Zone. All reads and writes flow through this instance during normal operation.



Synchronous Standby Replica

A second, identical instance is provisioned in a different AZ. Every transaction committed to the primary is *synchronously* replicated to the standby before the write is acknowledged.



Automatic Failover

If the primary instance fails, Amazon RDS automatically promotes the standby — typically within 60–120 seconds — with no manual intervention required.





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Task 5: Connect the Web Application to RDS

Now that your RDS instance is **Available**, you will copy its endpoint and use it to configure the Address Book web application – proving your entire infrastructure is wired together correctly.

Connecting the App — Step by Step

→ Copy the RDS Endpoint

In the RDS Console, click on **lab-db**. Under the **Connectivity & security** tab, find the **Endpoint** string (e.g., `lab-db.c1234567890.us-east-1.rds.amazonaws.com`). Copy it to your clipboard.

→ Open the Web Application

Open a new browser tab, paste the Public IPv4 address into the address bar, and press Enter. The **Address Book** application interface will load.

→ Get the Web Server's Public IP

Navigate to **EC2** → **Instances**. Select the **Web Server** checkbox. In the Details pane below, copy the **Public IPv4 address**.

→ Configure the DB Connection

Select the **RDS** or **Configuration** tab in the app. Enter: **Endpoint** = your RDS endpoint, **Database** = `lab`, **Username** = `main`, **Password** = `lab-password`. Click **Submit**.

Environment Checkpoints & Validation

Before submitting your lab for grading, confirm each checkpoint below. All four must pass for full credit.

VPC Security Group

Inspect DBSG in the VPC Console. Confirm the inbound rule shows **Port 3306** with source set to **Web-Security-Group**.

Multi-AZ Architecture

Under lab-db's **Connectivity & security** tab in RDS, confirm the primary and standby instances reside in **different Availability Zones**.

Database Connectivity

The web application displays a **"Connected to RDS DB"** status banner after submitting connection credentials.

Data Persistence

Create a sample contact (e.g., *Jane Doe*) in the Address Book. Refresh the browser and confirm the record still appears – proving the database is writing and reading correctly.

Submitting for Automated Grading

Once all four checkpoints pass, submit your work. The AWS Academy platform will run a background validation script against your deployed resources.

01

Return to the Lab Instructions Panel

Switch back to the primary browser window showing the AWS Academy lab instructions — not the AWS Console or the web application.

03

Wait for Validation

Allow **1 to 2 minutes** for the background grading script to evaluate your deployed resources. Do not close the window or restart the lab during this period.

02

Click Submit

At the top of the lab interface, select the **Submit** button. When prompted with a confirmation dialog, click **Yes** to confirm your submission.

04

Review Your Score

Click **Grades** or view the **Submission Report** to see your score breakdown by task. Each checkpoint corresponds to an automated check.



Troubleshooting Guide

Running into problems? Use this matrix to diagnose and fix the most common issues in Lab 5. Each symptom maps to a root cause and a clear remediation path.

Common Errors & Fixes

Symptom	Root Cause	Remediation
Web App: "Connection Timeout"	DBSG inbound rule missing Port 3306, or wrong source SG selected.	VPC Console → Security Groups → DBSG → Edit Inbound Rules. Verify Type = MySQL/Aurora (3306) and Source = Web-Security-Group. Save and retry.
Status stays "Creating" >15 min	RDS is provisioning across multiple AZs during high classroom concurrency.	Do not delete or restart the job. Allow up to 15 full minutes. Use the RDS console's internal refresh button to poll status.
"Access Denied for user 'main'"	Incorrect credentials entered in the web application config form.	Re-enter carefully: Username = <code>main</code> , Password = <code>lab-password</code> , Database = <code>lab</code> . Credentials are case-sensitive.
Cannot select subnets in Subnet Group	Subnets from the default VPC selected instead of Lab VPC private subnets.	Delete the subnet group. Re-create it with VPC = Lab VPC and select private subnets 10.0.1.0/24 and 10.0.3.0/24.
Console: "Access Denied" on DB create	Unauthorized engine, template, or paid feature selected (blocked by Academy SCPs).	Use MySQL, Dev/Test template, db.t3.micro, and uncheck Performance Insights and Enhanced Monitoring.

Key Concepts Recap

Before you leave, make sure these foundational cloud concepts are clear — they underpin everything you built in this lab.



Principle of Least Privilege

Security Groups grant only the minimum access required. DBSG allows Port 3306 from one source only — the web tier.



Multi-AZ High Availability

By placing the primary and standby in separate AZs, RDS survives a single-zone failure with automatic failover and no data loss.



Private Database Placement

Setting Public Access = No and using private subnets ensures the database is never reachable from the internet — only from within the VPC.



Managed Service Benefits

Amazon RDS handles OS patching, backups, and replication automatically — freeing you to focus on application logic rather than infrastructure operations.

Lab Complete — Well Done!

You've successfully deployed a **production-style, Multi-AZ relational database infrastructure** on AWS — from scratch, entirely through the console.

4

Tasks Completed

Console access, Security Group, Subnet Group, RDS Instance + App integration

2

Availability Zones

Your database spans two AZs with automatic failover protection

0

Servers Managed

Amazon RDS handles all infrastructure — you just configure and connect

You are now ready to tackle more advanced AWS database, networking, and application integration scenarios. Great work! 🎉

